Competitive Exam Practice 12: Instruction Pipelining and RISC vs. CISC

GATE (Computer Science) Pattern

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September 5, 2026

Each question carries 1–2 marks. MCQ = single correct option; MSQ = multiple-select; NAT = Numerical Answer Type. Every item below is *[Original, GATE-pattern]*. No item is labelled as a real PYQ because the local scanned PYQ pages did not yield a complete, independently verifiable stem and answer.

Q1 *[Original, GATE-pattern]* [**NAT, 2 marks**] Twelve independent instructions execute in an ideal balanced five-stage pipeline with one cycle per stage and no stalls. What are the pipelined completion time and the finite-stream speedup over a non-pipelined execution, respectively? Report the speedup to two decimal places.

Q2 *[Original, GATE-pattern]* [**NAT, 2 marks**] Consider the five-stage pipeline `IF, ID, EX, MEM, WB` and the sequence `LW R1,0(R2)`, `ADD R3,R1,R4`, and `SUB R5,R3,R6`. Load data is available after MEM; ALU results are forwarded at the end of EX. Starting the first IF in cycle 1, give the cycle in which `SUB` completes WB and the number of bubbles required, in the form "cycle, bubbles".

Q3 *[Original, GATE-pattern]* [**MCQ, 1 mark**] An ideal five-stage pipeline is full and has no hazards. Which statement is correct for a long independent instruction stream?

(A) Each instruction has one-cycle latency and one-cycle throughput.

(B) Each instruction has approximately five-cycle latency, while the stream can complete one instruction per cycle.

(C) Both latency and throughput are approximately five cycles per instruction.

(D) Throughput improves only if the pipeline has one stage.

Q4 *[Original, GATE-pattern]* [**NAT, 1 mark**] Branches are 20% of the dynamic instruction stream. The predictor mispredicts 10% of branches, and each misprediction costs 3 cycles. Ignoring all other effects, what is the branch contribution to CPI? Report to two decimal places.

Q5 *[Original, GATE-pattern]* [**MCQ, 1 mark**] In a five-stage pipeline, an instruction in MEM and a younger instruction in IF request the same single-ported unified memory in the same cycle. Which pair is the most direct classification and one valid remedy?

(A) Data hazard; forward the producer's ALU result

(B) Structural hazard; split instruction and data memories

(C) Control hazard; rename the destination register

(D) Data hazard; add a branch-prediction entry

Q6 *[Original, GATE-pattern]* **[MSQ, 1 mark]** Which statements are correct about the RISC/CISC comparison in a fixed workload?

(A) RISC often favors regular formats and register-centered execution.

(B) CISC can express more work per architectural instruction.

(C) RISC always has fewer instructions and CISC always has lower CPI.

(D) Total time should be compared using instruction count, CPI, and clock rate, rather than an ISA label alone.

Q7 *[Original, GATE-pattern]* **[MCQ, 1 mark]** For the same workload, a RISC-like implementation executes 1.20×10^9 instructions at CPI $= 1.1$ and $f = 2.4$ GHz. A CISC-like implementation executes 0.80×10^9 instructions at CPI $= 2.0$ and $f = 3.0$ GHz. Which statement is correct?

(A) RISC-like is faster because its CPI is lower.

(B) CISC-like is faster because its total time is approximately 0.533 s versus 0.550 s.

(C) Both take the same time because CPI is the only relevant metric.

(D) No comparison is possible without knowing the instruction names.

Q8 *[Original, GATE-pattern]* **[MSQ, 1 mark]** Which statements correctly identify parallelism or workload constraints?

(A) One instruction applied to eight data elements is SIMD-style data parallelism.

(B) Independent cores executing different tiles are MIMD-style parallelism.

(C) A large working set can make cache and virtual-memory behavior relevant even when the front end is pipelined.

(D) A higher clock rate alone proves that an implementation is faster for every workload.